

Components of an Information Retrieval System

Introduction

Nuno Escudeiro (nfe@isep.ipp.pt)

Ricardo Almeida (ral@isep.ipp.pt)

Session outline

1. Information Retrieval
2. Components of an Information Retrieval System
3. Retrieval Models
4. Evaluation and Performance Metrics
5. Practical Example
6. Future Directions and Challenges

1. Information Retrieval

Review.

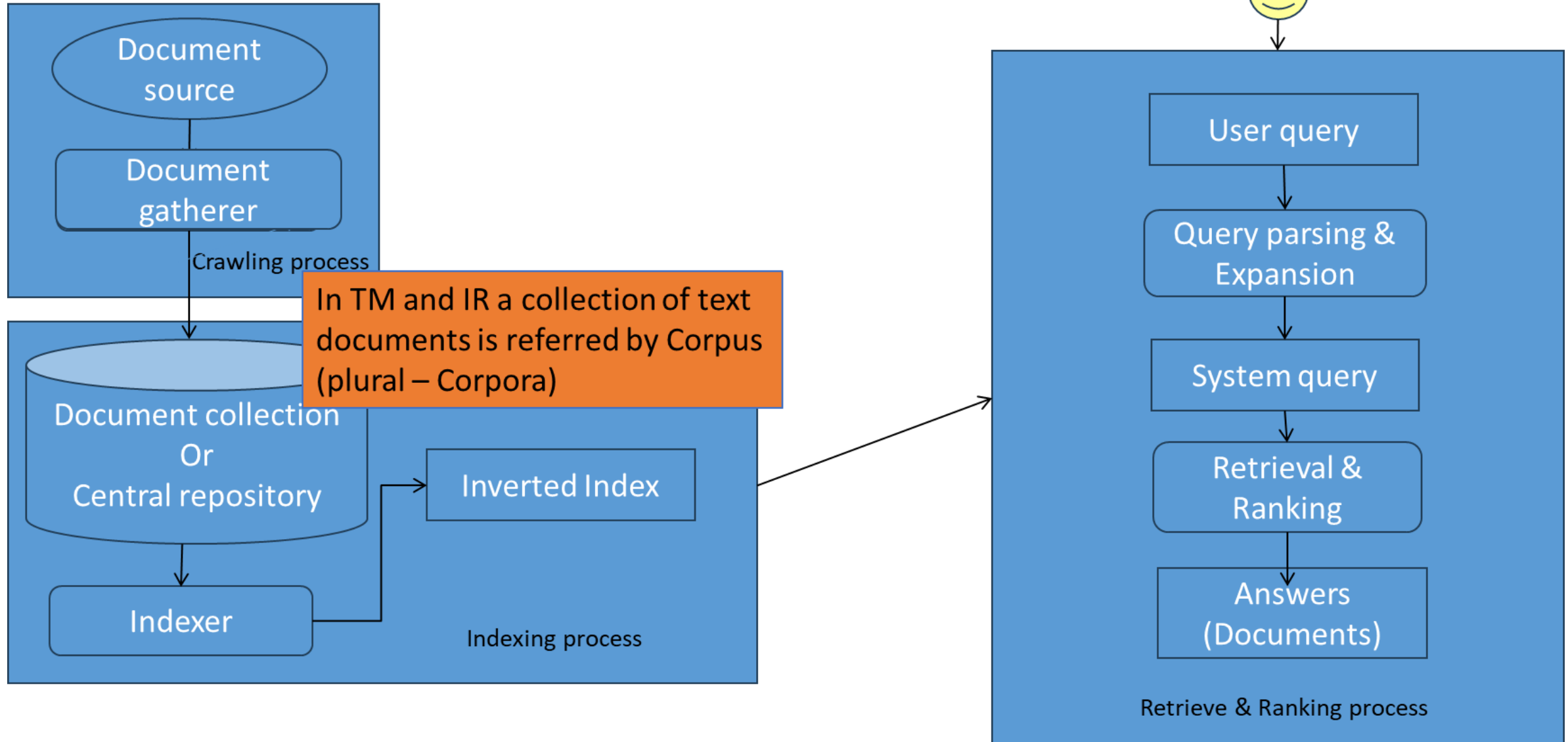
Information Retrieval: what is it?

- Information retrieval (IR) is the process of obtaining/finding relevant material/information according to the users' needs from large collections normally stored on computers.

2. Components of an IR System

Review.

Components of an IR System



3. Retrieval Models

Review.

Retrieval Models

- Boolean Models
- Vector Space Model (VSM)
- Probabilistic Models
- Language Models
- Latent Semantic Indexing (LSI) / Latent Semantic Analysis (LSA)
- Deep Learning Models

4. Evaluation and Performance Metrics

Review.

Evaluation and performance metrics

- Precision
- Recall
- F_β Score, F1 Score
- Mean Average Precision (MAP)
- Mean Reciprocal Rank (MRR)
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5. Practical Example

Review.

Evaluation and performance metrics

1. Open a web browser and make the following query: “web information retrieval”. Consider the answers retrieved in terms of what you expected to obtain from the query and evaluate the results obtained.
2. Compute precision and recall applied to the first 30 documents retrieved.
3. Compare precision and recall between each other

Evaluation and performance metrics

1.
 - Relevance of the retrieved documents:
 - **Precision** : number of retrieved docs, from all that were obtained, that are relevant to the user's information need
$$\text{Precision} = \frac{\text{Number of Relevant Documents Retrieved}}{\text{Total Number of Documents Retrieved}}$$
2.
 - **Recall** : number of relevant docs in collection that are retrieved
3.
$$\text{Recall} = \frac{\text{Number of Relevant Documents Retrieved}}{\text{Total Number of Relevant Documents in the Corpus}}$$

Information Retrieval

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6. Future Directions and Challenges

Review.

Future Directions and challenges

- Semantic search
- Personalized retrieval
- Multimedia/multimodal retrieval